INCH-POUND
MIL-DTL-24356B(NAVY)
23 October 2017
SUPERSEDING
MIL-C-24356A(SH)
21 April 1982

DETAIL SPECIFICATION

COUPLINGS AND SPLIT CLAMPS, 2½-, 4-, 6-, AND 7-INCH SEGMENTED, REATTACHABLE, FOR REFUELING-AT-SEA HOSE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 2½-, 4-, 6-, and 7-inch segmented, reattachable couplings and split clamps intended for use on 2½-, 4-, 6-, and 7-inch hoses meeting the requirements specified in MIL-H-22240 and primarily intended for use in, but not limited to, refueling at sea.

1.2 Classification. Couplings and split clamps covered by this specification are of the following sizes and types, as specified (see 6.2).

1.2.1 Sizes. Couplings and split clamps are of the following sizes (inside diameter of hose to be used):

a. 2½-inch
b. 4-inch
c. 6-inch
d. 7-inch

1.2.2 Types. Couplings and split clamps are of the following types:

a. Split clamp
b. Female coupling
c. Male coupling

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). PINs to be used for couplings and split clamps acquired to this specification are created as follows:

M	**24356**	**-**	**XX**	**XX**
Prefix for Military Specification	Specification Number	-	Size (see code below)	Type (see code below)

Size Code	
Size	**Code**
2½-inch	25
4-inch	40
6-inch	60
7-inch	70

Type Code	
Type	**Code**
Split clamp	SC
Female coupling	FC
Male coupling	MC

Example: M24356-25SC is for a 2½-inch split clamp.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys

MIL-H-22240 - Hose, Rubber, Petroleum Based Fuels and Water Services, Discharge Only Smooth Bore, Lightweight Buoyant Type

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

ASTM A376/A376M - Standard Specification for Seamless Austenitic Steel Pipe for High-Temperature Service

ASTM A630 - Standard Test Methods for Determination of Tin Coating Weights for Electrolytic Tin Plate

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ASTM B26/B26M	-	Standard Specification for Aluminum-Alloy Sand Castings
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B221	-	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B247	-	Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings
ASTM D2000	-	Standard Classification System for Rubber Products in Automotive Applications
ASTM E498/E498M	-	Standard Practice for Leaks Using the Mass Spectrometer Leak Detector or Residual Gas Analyzer in the Tracer Probe Mode
ASTM E515	-	Standard Practice for Leaks Using Bubble Emission Techniques
ASTM E1282	-	Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products
ASTM G69	-	Standard Test Method for Measurement of Corrosion Potentials of Aluminum Alloys

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS28775	-	Packing, Preformed – MS28775 O-Ring

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Material.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Material composition. The materials for all couplings and split clamps shall be as shown in table I.

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TABLE I. Material composition.

Name of Part	Figure Number	Material [1/]	Applicable Document	Remarks
Coupling nipple: male or female	1	Cast or wrought aluminum alloy	ASTM B26/B26M, Alloy 356-T6 or ASTM B221, 6061-T6	Anodized, MIL-A-8625, type I
Split clamp half	2	Aluminum wrought or extruded	ASTM B221, 6061-T6 or ASTM B247, 6061-T6	Anodized, MIL-A-8625, type I
Band for split clamp	3	Corrosion-resistant steel	ASTM A240/A240M	---
Screws and nuts	3	Corrosion-resistant steel	ASTM A276/A276M, type 316	---
End coupling body	1	Wrought aluminum alloy	ASTM B221, 6061-T6 or ASTM B247, 6061-T6	Anodized, MIL-A-8625, type I
O-ring	---	Synthetic rubber	ASTM D2000	SAE AS28775 for dimensions only

Coupling size (inches)	O-ring dash size
2½	234
4	346
6	438
7	442

NOTE:
[1/] Measured at valve stem or manual actuator stem.

3.3 Design.

3.3.1 Clamp design. The dimensions of the male and female couplings shall be in accordance with figure 1. The dimensions of the split clamp and band assembly shall be in accordance with figures 2 and 3. The female clamp coupling shall be fitted with an O-ring seal in the groove. Split clamp designs other than those shown on figures 2 and 3 are acceptable subject to approval of the Naval Sea Systems Command (NAVSEA). However, the proposed split clamp design shall be interchangeable with any couplings which can be obtained under this specification. The split clamp shall not have parts which can be lost during assembly or disassembly. Designs currently being used by the Fleet are displayed on figure 4.

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NOTES:
1. ALL DIAMETERS MUST BE CONCENTRIC WITHIN 0.010" TIR.
2. TOLERANCES SHALL BE ±.005 ON DECIMAL DIM. $\pm\frac{1}{64}$ ON FRACTIONS WHERE NOT SPECIFIED.
3. DETAILS NOT SPECIFIED SHALL BE AS DESIGNATED BY THE CONTRACTOR AND AS APPROVED BY NAVSEA.
4. ALL GROOVE SURFACES TO BE 63.

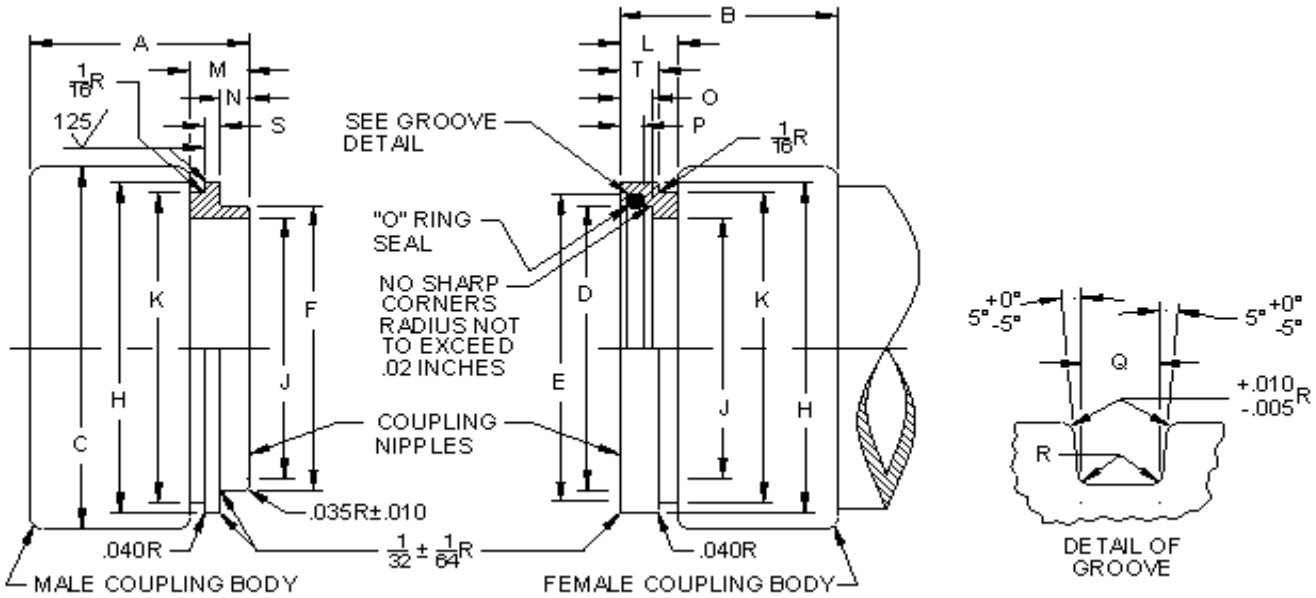


FIGURE 1. End coupling body – male and female.

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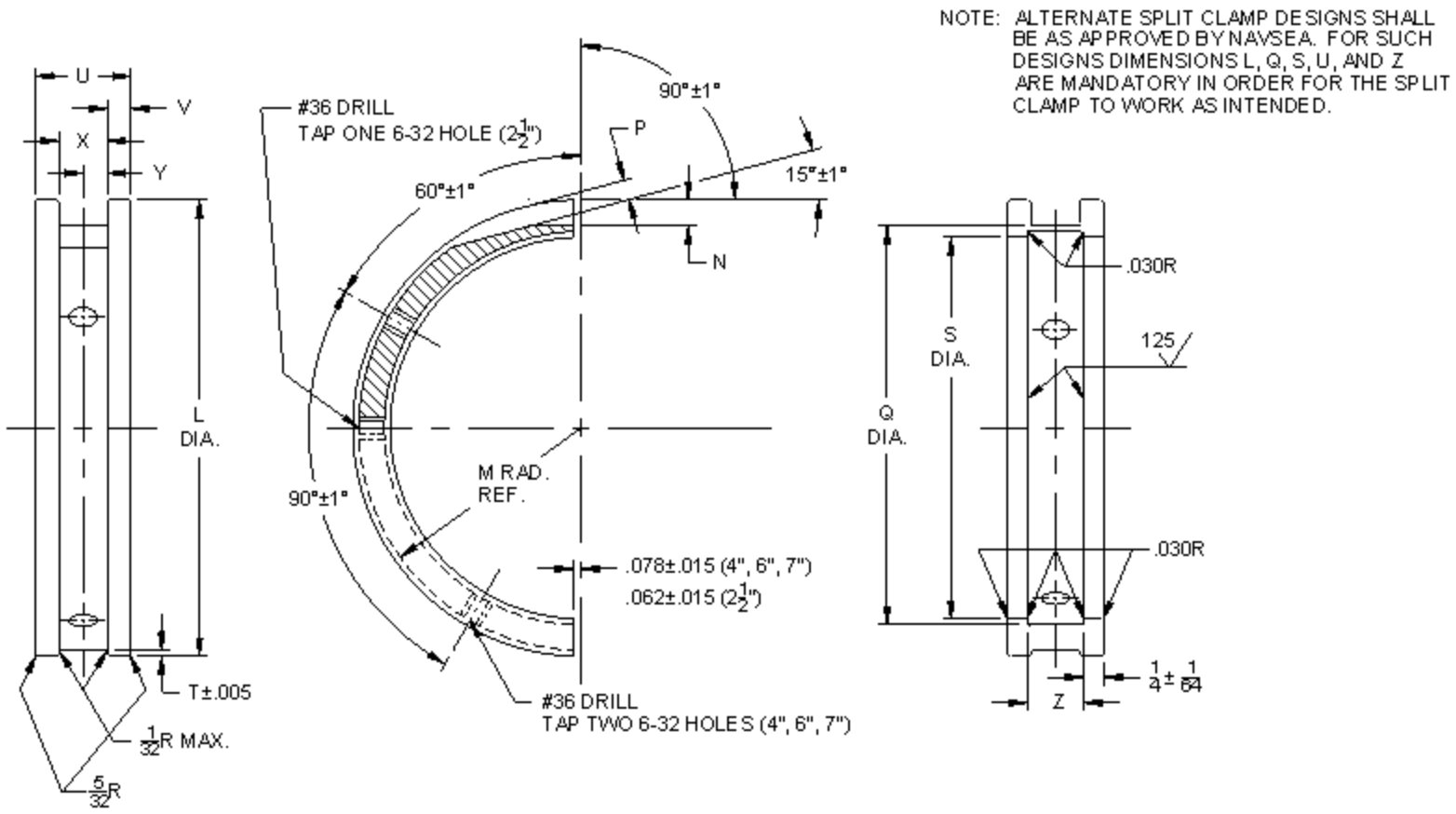
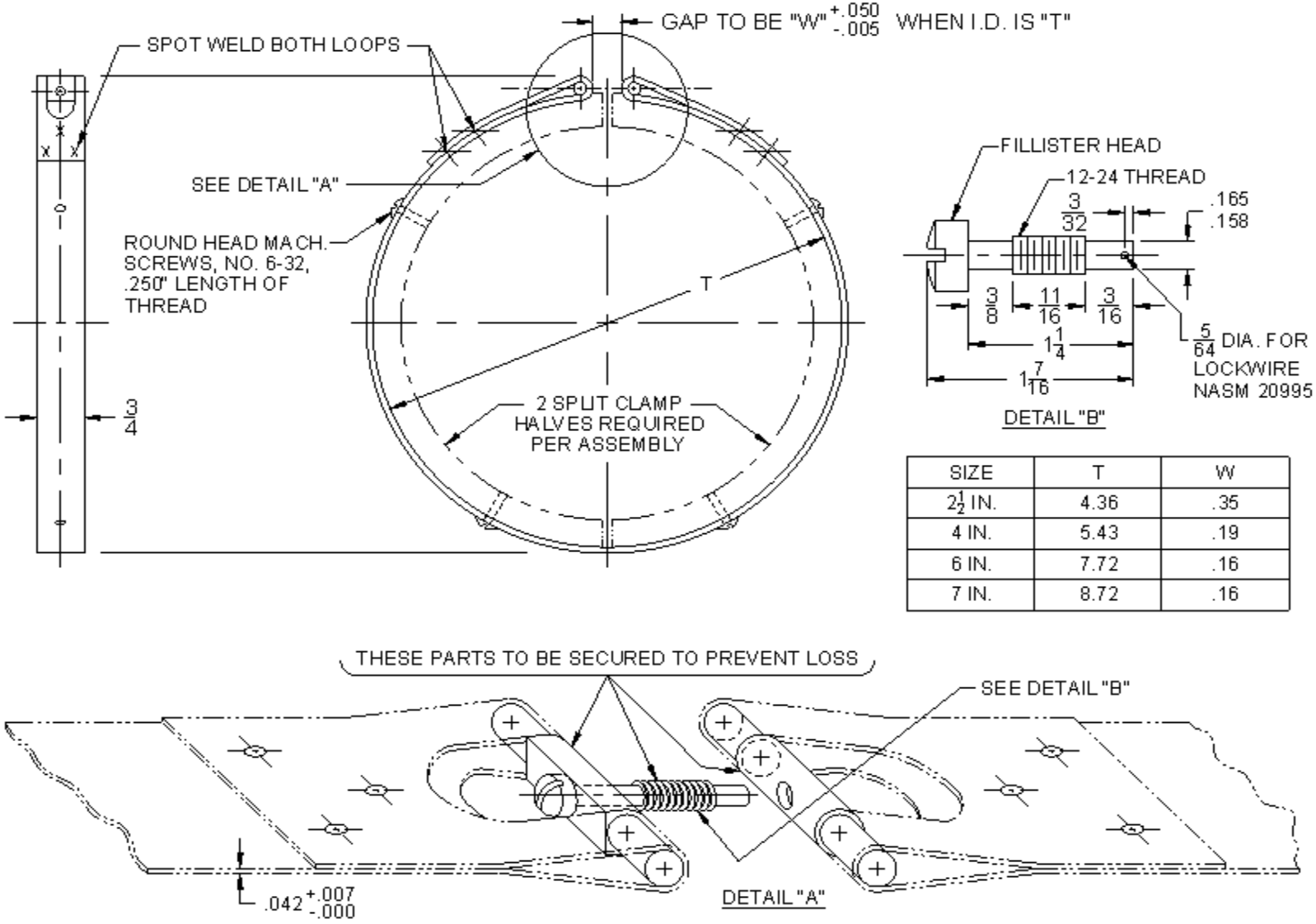


FIGURE 2. Split clamp half – details.

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SIZE	T	W
2 1/2 IN.	4.36	.35
4 IN.	5.43	.19
6 IN.	7.72	.16
7 IN.	8.72	.16

FIGURE 3. Split clamp and band – assembly and details.

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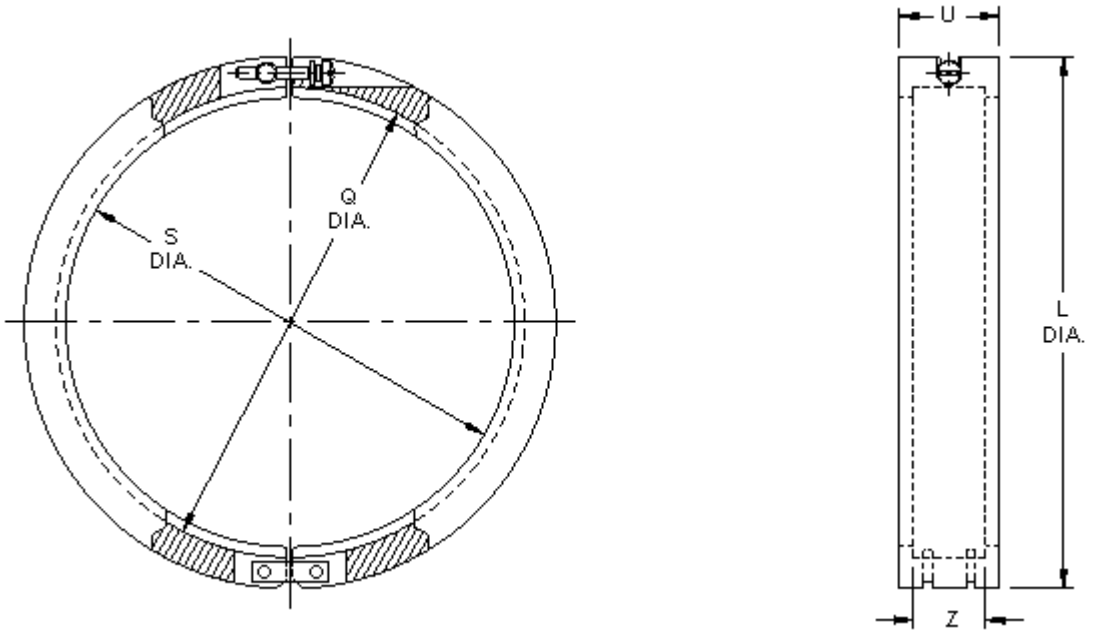


FIGURE 4. New split clamp – details.

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3.3.2 End coupling design. The end couplings shall be of the segmented, reattachable type securely gripping the hose, constructed as specified in MIL-H-22240, and maintaining such grip without significant relaxation. The internal nipple and external members in contact with the hose shall have the largest possible radii at their ends to prevent damage to the hose as a result of severe flexing or collapsing of the hose. The end of the nipple inside the hose shall be not more than ½ inch back from the end of the external clamping member. The coupling exterior shall be free of projections which can snag on ship's structure or become damaged through rough handling.

3.3.2.1 Design configuration. The 2½- and 4-inch end couplings may be a solid socket design.

3.3.3 Assembly/disassembly. The coupling shall be readily removable and reattachable to a new section of hose by the use of commonly available hand tools or tools which are normally carried onboard ship. The latter should be verified with the contracting activity at the time that approval is requested. The coupling should incorporate means to grip the hose to its predetermined, optimum amount only and to ensure that this condition can be repeated by any operator without exercising special skill or judgement. It shall be readily evident to the operator that the coupling is properly constructed. Conversely, means shall be incorporated to prevent improper coupling construction. The coupling design shall be such that the grip on the hose will not tend to decrease with a pull on the hose.

3.4 Performance requirements. The hose couplings shall meet all the applicable performance requirements specified herein when attached to hose constructed as specified in MIL-H-22240 without developing any weakness and without causing damage to the hose. (In this connection, reasonable deformation of the hose in the coupling shall be permitted, provided no damage or incipient failure is created thereby.) The gripping action shall involve the reinforcement without the deformation causing any damage to any part of the hose; reliance shall not be placed upon gripping of the tube and cover alone. All components of the coupling shall be designed to withstand rough handling, including dropping, without deformation.

3.4.1 Tensile pull. Couplings, when attached to a hose meeting the requirements specified in MIL-H-22240, shall withstand the tensile pull test specified in 4.7.2 without any visible damage to the hose or weakness of the coupling grip.

3.4.2 High pressure hold. Couplings, when attached to a hose meeting the requirements specified in MIL-H-22240, shall withstand the high pressure hold test specified in 4.7.3 without any indication of malfunction of leakage of fittings.

3.4.3 Proof pressure. Couplings, when attached to a hose meeting the requirements specified in MIL-H-22240, shall pass the proof pressure test specified in 4.7.4 without any leakage, deformation of hose coupling, or coupling blow off.

3.4.4 Rough handling. Couplings, when attached to a hose meeting the requirements specified in MIL-H-22240, shall withstand the drop test specified in 4.7.5 without any malfunction, leakage, physical damage, or weakness of the coupling components.

3.5 Marking. Each coupling body and segment shall be stamped at a place so as not to weaken the coupling. Stamping shall include, but not necessarily be limited to, the following:

a. Manufacturer's name or trademark.

b. Size, in inches.

c. Specification number (MIL-DTL-24356B).

3.6 Workmanship. Workmanship shall be such that there are no sharp or rough edges that could damage the hose or injure personnel. All parts shall be free of dirt, metal shavings, or other contamination.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on reattachable end couplings when a first article sample is required (see 3.1). First article inspection shall include the examinations of 4.4 and tests specified in table II.

4.2.1 Samples for first article inspection.

4.2.1.1 Reattachable end couplings. Four sets of reattachable end couplings of each size shall be subjected to the examinations of 4.4 and tests specified in table II. Each reattachable end coupling shall be supplied with a mating test plug secured with a split clamp assembly (see 4.7).

TABLE II. First article inspection.

Test No.	Test	Requirement	Verification	First Article Test Sequence 1/			
				Sample Number			
				1	2	3	4
1	Hose coupling assembly	3.3.3	4.7.1	1, 5	1, 5	1, 4	1, 6
2	High pressure hold test	3.4.2	4.7.3	3	3	-	-
3	Proof pressure test	3.4.3	4.7.4	2, 6	2, 6, 8	2, 5, 7	2, 4, 7
4	Tensile pull test	3.4.1	4.7.2	-	7	6	-
5	Drop test	3.4.4	4.7.5	-	-	-	3
6	Hose coupling disassembly 2/	3.3.3	4.7.1	4	4	3	5

NOTES:

1/ For each first article sample, tests shall be conducted in the numerical order represented by the numbers in the blocks.

2/ After disassembly, it shall be verified that the hose is not damaged. Then, 6 inches at each end of the hose shall be cut off and Test No. 1 shall be repeated. Elapsed time from completion of Test No. 6 to rerun of Test No. 1 on shortened sample shall be no more than 1 hour.

4.2.1.2 Hose. Hose samples required for first article testing of reattachable end couplings shall be as specified in MIL-H-22240. Wall thicknesses shall be as specified in table III. Only the cover of the hose samples shall be ground, if necessary, to obtain the desired wall thickness.

TABLE III. Dimensions of hose samples.

Sample Number	Minimum Length of Hose Between Couplings (Feet)	Wall Thickness (Inch) 1/, 2/
1	2	0.454 min.
2	2	0.410 max.
3	2	0.454 min.
4	2	0.410 max.

NOTES:

1/ The maximum and minimum dimensions required for the hose samples shall be evident on at least 80 percent of the hose circumference.

2/ Where 0.454-inch minimum wall thickness is required for testing, the actual maximum wall thickness shall not exceed that specified in MIL-H-22240. Where 0.410-inch maximum wall thickness is required for testing, the actual minimum wall thickness shall be not less than that specified in MIL-H-22240.

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4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests specified in table IV.

TABLE IV. Conformance inspection.

Test	Requirement	Verification
Hose coupling assembly	3.3.3	4.7.1
Hose coupling disassembly	3.3.3	4.7.1

4.3.1 Sampling for conformance inspection.

4.3.1.1 Lot. A lot shall consist of all reattachable end couplings manufactured under the same conditions and of the same size offered for delivery at one time.

4.3.1.2 Sampling for visual and dimensional examination. A random sample of reattachable end couplings shall be selected from each lot in accordance with table V for the examinations specified in 4.4.

TABLE V. Sampling for visual and dimensional examination.

Number of Reattachable End Couplings in Lot	Number of Reattachable End Couplings in Sample
15 and under	10
16 to 40	15
41 to 110	25
111 to 300	35
301 and over	50

4.3.1.3 Sampling for tests. One set of reattachable end couplings shall be selected at random from each lot and shall be assembled with a length of hose, meeting the requirements as specified in MIL-H-22240, to provide a free length between couplings of not less than 36 inches or more than 40 inches and subjected to the tests specified in table IV.

4.4 Examination. Each of the sample reattachable end couplings selected in accordance with 4.2.1.1 and 4.3.1.2 shall be examined to verify conformance to the requirements of this specification with respect to dimensions and tolerances (see 3.3.1 and 3.3.2), finishes and materials (see 3.2), marking (see 3.5), and workmanship (see 3.6). Failure to meet any of the requirements specified shall be cause for rejection.

4.7 Test methods. Physical tests for metal parts shall be determined in accordance with ASTM E498/E498M, ASTM E515, ASTM A630, ASTM A376/A376M, ASTM B117, ASTM G69, ASTM G47, and ASTM E1282 as applicable. For tests which require couplings attached to a section of hose, each end coupling shall have a mating test plug connected thereto by a split clamp assembly. The outside diameter of the mating test plugs shall not exceed the inside diameter of the coupling bodies. Couplings shall be installed on hose for those tests requiring hose assemblies.

4.7.1 Assembly/disassembly. The coupling assembly shall be easily assembled and disassembled using shipboard available tools and not damage the hose. Failure to meet the requirements of 3.3.3 shall be cause for rejection.

4.7.2 Tensile pull test. The test hose assemblies shall be subjected to the pull specified in table VI. The pull shall be maintained for at least 5 minutes. Failure to meet the requirements of 3.4.1 shall be cause for rejection.

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TABLE VI. Tensile pull.

Hose Size (Inches)	Tensile Pull (Pounds)
2½	5,000
4	5,000
6	12,000
7	12,000
6 (MIL-H-22240, type B)	16,000

4.7.3 High pressure hold test. The sample for the high pressure hold test shall have the ends blanked off and shall be subjected to a hydrostatic pressure equal to 70 percent of the minimum burst pressure for a period of 5 minutes. The pressure shall then be dropped to zero pounds per square inch (lb/in^2). The assembly shall then be repressurized to the above pressure for an additional 5 minutes. Failure to meet the requirements of 3.4.2 shall be cause for rejection.

4.7.4 Proof pressure test. Hose couplings shall be subjected to the proof pressure specified in table VII for not less than 5 minutes. Failure to meet the requirements of 3.4.3 shall be cause for rejection.

TABLE VII. Proof pressure.

Hose Coupling (Inches)	Proof Pressure (lb/in^2)
2½	300
4	300
6	300
7	250

4.7.5 Rough handling. Hose couplings shall be dropped five times onto a steel or set concrete floor from a height of 10 feet, such that both end couplings hit at approximately the same time. The mating test plugs (see 4.7) shall remain in the couplings during the drop test specified in table II. Failure to meet the requirements of 3.4.4 shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Coupling assemblies covered by this specification are intended for use with hose as specified in MIL-H-22240 in fueling-at-sea applications aboard naval ships. The maximum operating pressure of fueling-at-sea systems is 150 lb/in^2. Existing stock of couplings and split clamps manufactured to MIL-C-24356A, Amendment 2, dated 1 March 1988, are acceptable for use until depleted.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable size(s) and type(s) (see 1.2).

c. PIN required (see 1.3).

d. When first article inspection is required (see 3.1 and 6.3).

e. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required (see 6.2), the item should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Coupling body

Female coupling

Male coupling

Refuel-at-sea

Split coupling

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
Navy – YD

Preparing activity:
Navy – SH
(Project 4730-2017-095)

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